

Lec-22

Lecture after Quiz-3

→ HETERO Junctions

Hetero structure devices: devices with a position dependent alloy composition

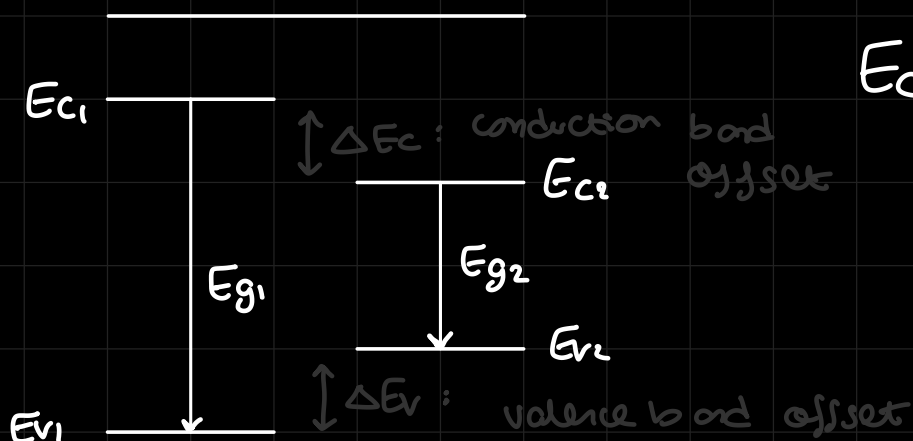
→ Type of hetero junctions

Straddling Gap

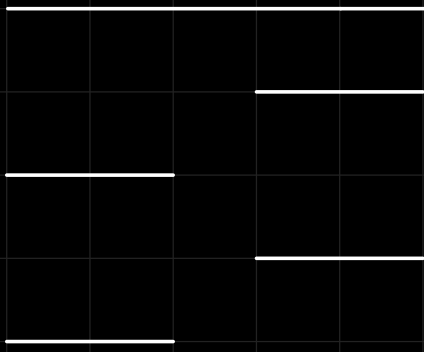
Staggered gap

Broken gap

① ~~Staggered~~ gap (Straddling)



(2) Staggered Gap



$$E_{c1} < E_{c2}$$

$$E_{v1} < E_{v2}$$

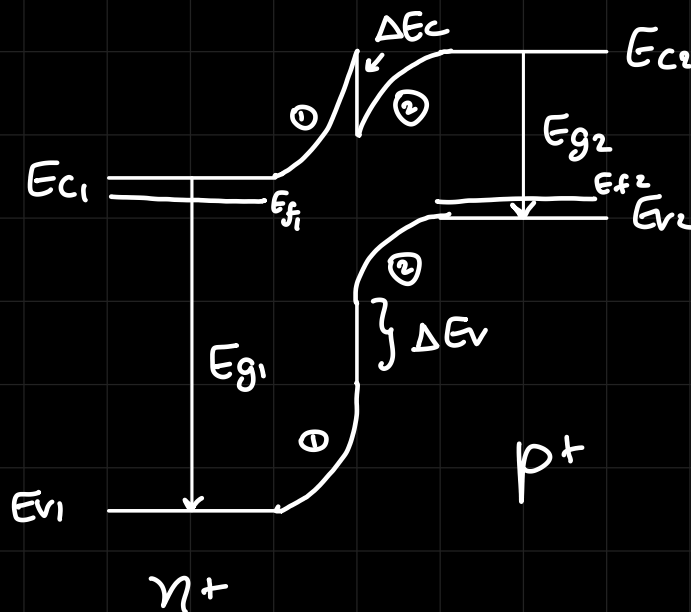
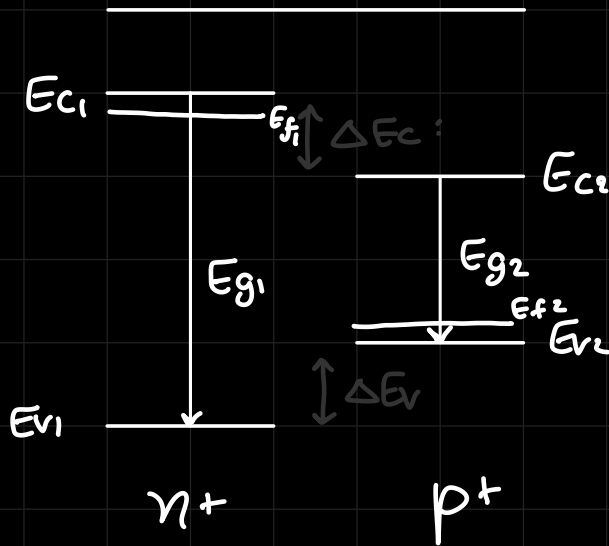
(3) Broken Gap



$$E_{c2} > E_{v2} > E_{c1} > E_{v1}$$

eg: $\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$

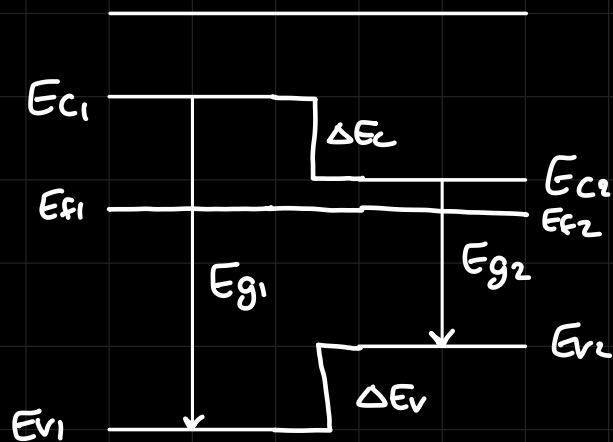
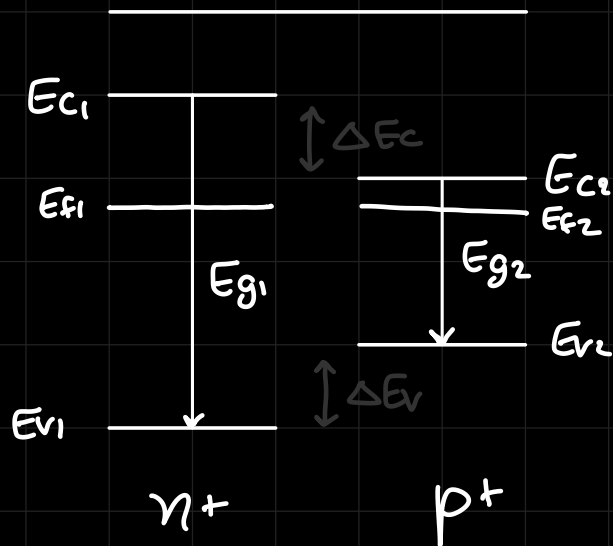
before contact



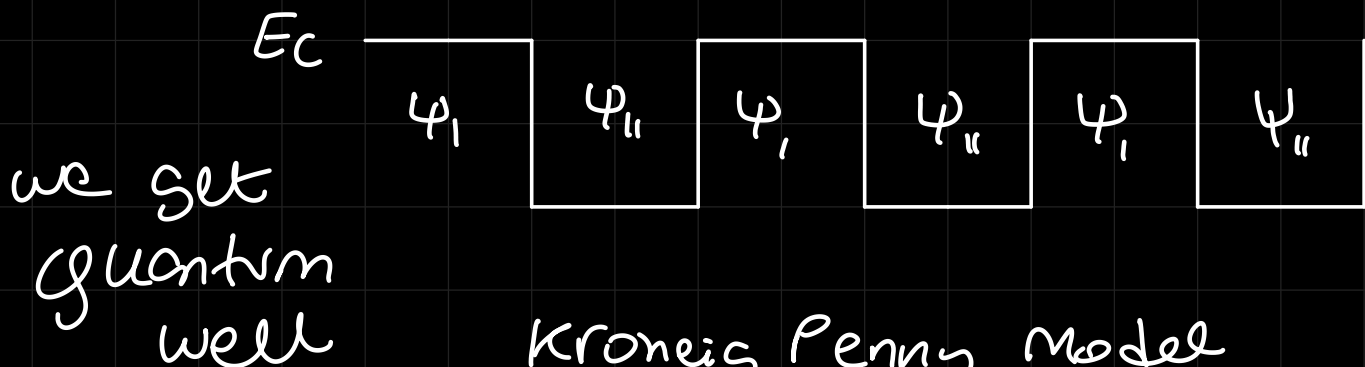
① : e^- depleted
② : e^- accumulated

how to determine $\bar{E}$ for the sudden jump due to the offsets?

How about we choose doping so that
no band bending required (E_F const)



What if we add more such SC pairs?



Kroneig Penny Model
quantized periodicity

This conduction band will form minibands. We can also think about valence band similarly.

miniband (super lattice design)

quantum cascade laser



if the distance of traversal is kept constant, we can achieve a radiation of same freq with very high intensity (LASER)

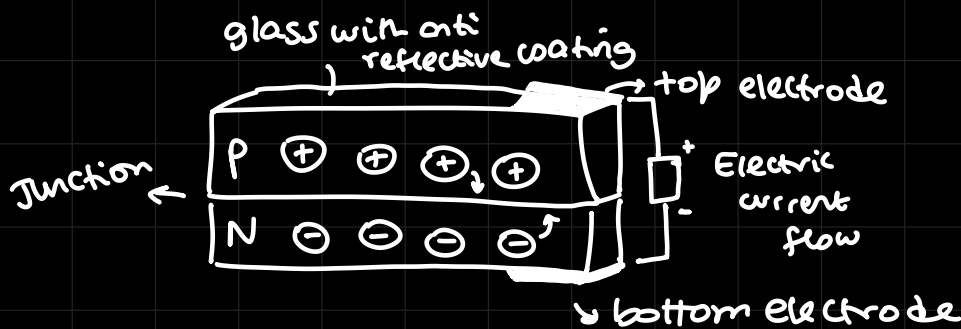
SOLAR CELL

$h\nu$ absorbed : $e^- \rightarrow$ free electron hole pair generated

Utilizes a PN Junction

↓
This charge is held in a solar cell

works in unbiased mode



The charges in the depletion region must be immediately separated ($h\nu$) and stored

$$I = I_{ph} - I_d$$

photogenerated current

like leakage current
diode/dark current

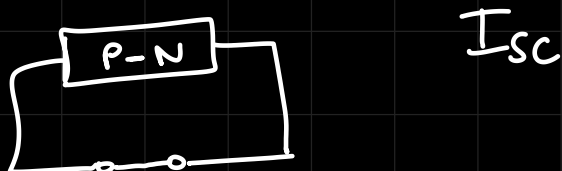
due to thermal excitation inside the device

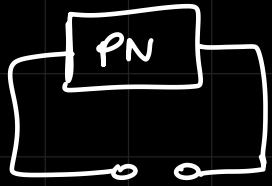
need to minimize I_d and maximize I_{ph}

$$I = I_{ph} - I_0 \left(e^{\frac{qV}{k_B T}} - 1 \right)$$

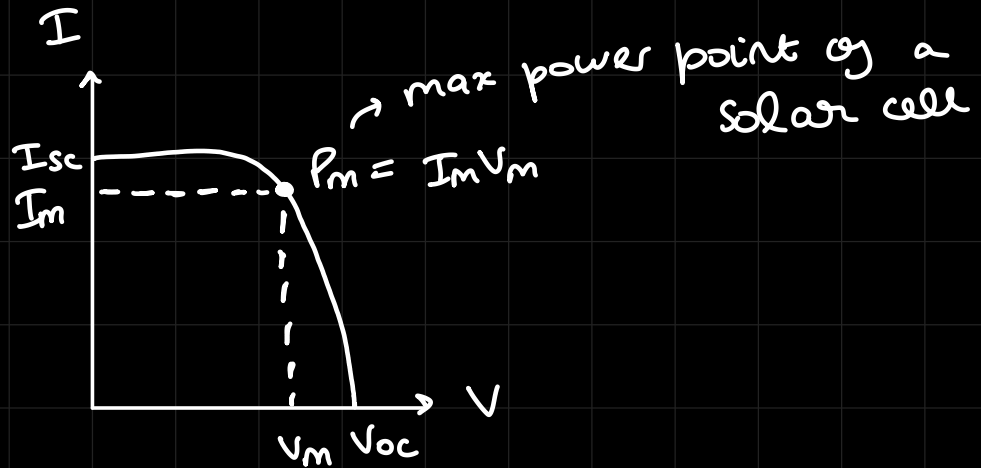
* Short circuit current : max I generated

with no bias/resistor





* Open ckt voltage : max V generated
(V_{oc}) in open ckt condition



SOLAR efficiency (Shockley - Queisser Limit)

The max efficiency for a single
pn junction solar cell is 33.7%
at $E_g = \underline{1.4\text{ eV}}$

but Si $\rightarrow E_g = 1.1\text{ eV}$
resulting in $\eta_{\text{max}} = \underline{32\%}$ only

visible range : $1.6\text{ eV} \rightarrow 3.4\text{ eV}$

but $\eta_{33.7} \rightarrow E_g = 1.4\text{ eV} < 1.6\text{ eV}$

We get better η efficiency by stacking layers of diff materials (with varying band gaps) to capture wider range of frequency spectrum.

$\equiv$ Tandem Solar Cell

Theoretically, we got 50% - 52% η

Still we use Si because practical use case wise, it offers mix of η and complexity.